

PERSONAL INFORMATION

Mohammad Mehedi Hasan Munna



 Dhaka, Bangladesh

 +8801580660139

 mohammadmehedi.munna@gmail.com

 www.linkedin.com/in/mohammad-mehedi-hasan-munna-719a03225/

 [ORCID 0009-0000-0860-9337](https://orcid.org/0009-0000-0860-9337)

 [Google Scholar 1kIGglAAAAAJ](https://scholar.google.com/citations?user=1kIGglAAAAAJ)

Date of birth 1 March 2003 | **Nationality** Bangladeshi

RESEARCH PROFILE

Computer science graduate working on machine learning for medical image analysis, with a focus on label-efficient and self-supervised methods for settings where annotated clinical data is scarce. Undergraduate thesis develops self-supervised MRI reconstruction aimed at deployment in resource-constrained hospitals in Bangladesh. Author of one journal article, nine peer-reviewed conference papers and one Springer book chapter, with first authorship on seven.

EDUCATION AND TRAINING

Feb 2022–Aug 2026

Bachelor of Science in Computer Science and Engineering

East West University, Dhaka, Bangladesh

- **Thesis:** Self-Supervised Learning for Offline MRI Reconstruction: Experimental Analysis Towards Clinical Use in Bangladesh. Supervisor: [ACTION: add supervisor name].
- **Mathematics coursework:** [ACTION: list Linear Algebra, Probability & Statistics, Calculus, Discrete Mathematics, Numerical Methods etc. exactly as on the transcript, with credit hours. Several Erasmus Mundus programmes set a minimum ECTS volume in mathematics — this line decides eligibility, so do not leave it vague.]
- **Computing coursework:** Artificial Intelligence, Machine Learning, Digital Image Processing, Computer Vision, Data Structures & Algorithms, Statistics for Data Science.
- **CGPA:** 3.17/4.00.

RESEARCH INTERESTS

Medical image analysis and reconstruction • Self-supervised and label-efficient representation learning • Computer vision for clinical and low-resource settings • Trustworthy and explainable AI for healthcare

RESEARCH EXPERIENCE

Dec 2025–May 2026

Research Assistant (Intern)

Center for Computational & Data Sciences (CCDS), Independent University Bangladesh

- Developed self-supervised computer vision methods for medical image segmentation under limited annotation budgets.
- Built automated annotation pipelines for agricultural imaging datasets, including a tea leaf disease corpus.
- **Tools:** Python, PyTorch, YOLOv8, OpenCV.

Jan 2025–Jun 2025

Tech Course Analyst & Coordinator (Intern)

ICT Olympiad Bangladesh, Dhaka, Bangladesh

- Redesigned ICT curriculum content to align with national and international standards.
- Surveyed emerging technology trends to inform course development.

PUBLICATIONS

Own name in **bold**. Full list at Google Scholar.

Journal Articles

- [J1] Jawad, M. A., **Munna, M. M. H.**, Kabir, A. H., Antu, N. H., & Tulona, R. F. (2026). A runtime safety copilot for AI-native O-RAN: Predictive verification and fail-safe enforcement in near-RT RIC control loops. *IEEE Access*. DOI: 10.1109/ACCESS.2026.3686132

Under Review

- [U1] **Munna, M. M. H.**, Chowdhury Rafi, A. M., Sami, M. J. I., Sifat, K. M., Sultana, T., UI Islam, R., Tulona, R. F., & Kabir, A. H. Self-supervised learning for accelerated MRI: A critical review of paradigms, trustworthiness, and translational challenges. *Journal of Magnetic Resonance Imaging (JMRI)*. Under review, submitted [ACTION: Month Year]. *First and co-corresponding author*.
- [U2] **Munna, M. M. H.**, Islam, T., Faruk, O., Tulona, R. F., Sultana, A., Saima, F., Kabir, A. H., Sultana, T., & UI Islam, R. Artificial intelligence for women's reproductive health: A scoping review of global diagnostic trends, methodological gaps, and a translational research agenda for low-resource settings. *PLOS ONE*. **Major revision (R1) under review**, submitted [ACTION: Month Year]. *First and corresponding author*.
- [U3] **Munna, M. M. H.**, Aman, N., Kabir, A. H., Tulona, R. F., Saha, S., Kabir, S. S., Habib, I., Dhruvo, V., & Sultana, T. JPEG2000 compression thresholds for deep learning-based cancer histopathology classification: A multi-dataset, multi-tissue evaluation with compression-aware training. *PLOS ONE*. Under review, submitted [ACTION: Month Year]. *First and corresponding author*.
- [U4] **Munna, M. M. H.**, Tulona, R. F., & Kabir, A. H. AGRI-KD: A lightweight multi-task framework for rice disease diagnosis with localization and severity quantification. *Smart Agricultural Technology*. Under review, submitted [ACTION: Month Year]. *First and corresponding author*.

Two further co-authored manuscripts are under review at *IEEE Access* (IoT gateway architecture; embedded flood-risk sensing).

Book Chapters

- [B1] **Munna, M. M. H.**, Aman Turzo, N., Saha, S., Islam, T., Safy, I. H., Saima, F., & Nusrat, S. (2025). Explainable hybrid convolutional networks for accurate malaria diagnosis in blood smear images. In *Data Science, AI and Applications (ICDSAIA 2025)*, Communications in Computer and Information Science, vol. 2682. Springer, Cham. DOI: 10.1007/978-3-032-11352-8_4

Conferences: Medical Imaging & Health Informatics

- [C1] **Munna, M. M. H.**, Afrin Anamika, S., Sultana, A., Mia, M. S., Islam, T., Mahmud Bitto, P., Sultana, T., Rashid, M. R. A., Ul Islam, R., Reza, A. W., & Ripon, S. H. (2026). AI-assisted PCOS screening from ultrasound using vision transformers and explainable attention. In *5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*. DOI: 10.1109/ICECTE69292.2026.11429340
- [C2] **Munna, M. M. H.**, Aman, N., Kabir, S. S., Bari, S., Islam, T., Dhruvo, V., Kabir, A. H., & Tulona, R. F. (2025). Deep learning for reliable healthcare: Insights from chest X-ray prediction. In *28th International Conference on Computer and Information Technology (ICCIT)*. Accepted, oral presentation.
- [C3] **Munna, M. M. H.**, Aman, N., Bari, S., Mia, M. S., Kabir, A. H., Tulona, R. F., & Enan, I. I. (2025). Contrastive representation learning for acute lymphoblastic leukemia: A SimCLR and EfficientNet-B0 approach. In *International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*. IEEE. Presented 27 November 2025.
- [C4] **Munna, M. M. H.**, Habib, I., Turzo, N. A., Jahan, N., & Nusrat, S. (2024). Automated diagnosis of acute lymphoblastic leukemia leveraging EfficientNet and fastai. In *27th International Conference on Computer and Information Technology (ICCIT)* (pp. 233–238). IEEE. DOI: 10.1109/ICCIT64611.2024.11021950
- [C5] **Munna, M. M. H.**, Habib, I., & Aman, N. (2024). A novel hybrid Xception-DenseNet121 model for acute lymphoblastic leukemia classification. In *6th International Conference on Sustainable Technologies for Industry 5.0 (STI)* (pp. 1–6). IEEE. DOI: 10.1109/STI64222.2024.10951150
- [C6] Habib, I., **Munna, M. M. H.**, & Aman, N. (2024). Leveraging hybrid deep learning for effective multiclass blood cancer detection. In *International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)* (pp. 1–5). IEEE. DOI: 10.1109/ICRPSET64863.2024.10955897

Conferences: Networks, Security & Systems

- [C7] Jawad, M. A., **Munna, M. M. H.**, Tulona, R. F., Antu, N. H., & Kabir, A. H. (2026). A comparative survey of IoT platforms: Architecture, security, and emerging trends toward distributed intelligence. In *8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)*. Accepted, oral presentation.
- [C8] Jawad, M. A., **Munna, M. M. H.**, Antu, N. H., Tulona, R. F., Kabir, A. H., & Islam, A. (2025). Optimized bandwidth allocation in cellular networks using genetic algorithms and a historical stability index. In *International Conference on Intelligent Computing and Next Generation Networks (ICNGN)*. DOI: 10.1109/ICNGN67480.2025.11413736
- [C9] **Munna, M. M. H.**, Aman, N., Saha, S., Islam, T., Nusrat, S., & Das, S. K. (2025). Deep learning in cybersecurity: A hybrid CNN framework for multi-source malware detection. In *International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)* (pp. 1–6). IEEE. DOI: 10.1109/QPAIN66474.2025.11171945

ACADEMIC SERVICE

2025–Present

Reviewer and Sub-Reviewer

- **Journals:** IEEE Access ([N] reviews); IJECE ([N] reviews).
- **Conferences and workshops:** NeurIPS 2026; MICCAI 2026 ShapeMI Workshop.
- **Sub-reviewer:** Pattern Recognition (Elsevier), under the supervision of [ACTION: name].

PERSONAL SKILLS

Mother tongue Bangla

English IELTS Academic: [ACTION: overall band and section scores, test date]. Medium of instruction throughout undergraduate study.

Other languages	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C2	B2	C1	C2

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
[Common European Framework of Reference for Languages](#)

TECHNICAL & PROFESSIONAL SKILLS

Programming Languages	Python (primary), C, Java, SQL
Deep Learning	PyTorch, TensorFlow, fastai, lightly.ai, Hugging Face
Medical Imaging	OpenCV, scikit-image, SimpleITK, NumPy, MATLAB
Methods	Self-supervised learning (SimCLR), contrastive learning, transfer learning, image segmentation, explainable AI (Grad-CAM), vision transformers
Scientific Stack	NumPy, pandas, scikit-learn, SciPy, Matplotlib, seaborn
Tooling	Git/GitHub, Linux, L ^A T _E X, Jupyter

LEADERSHIP AND OUTREACH

March 2026–Present	Secretary IEEE Computer Society Bangladesh Chapter Secretariat & Student Activities Committee (SAC), Dhaka, Bangladesh – Support Executive Committee initiatives, maintain chapter records and manage the chapter's online presence. – Coordinate inter-university student programmes, workshops and seminars.
Sep 2025–Present	Ambassador International Computer Science Competition (ICSC), Heidelberg, Germany – Represent the competition at universities and student societies; organise information sessions and workshops.
May 2025–March 2026	Public Relations Executive IEEE CS Bangladesh Chapter Student Activities Committee (SAC), Dhaka, Bangladesh – Managed digital outreach and served as external communications liaison for committee events.

AWARDS & HONOURS

2025	Best Intern — ICT Olympiad Bangladesh, Season 2
------	--

NETWORKS & MEMBERSHIPS

Dec 2024–Present	IEEE Student Member
Dec 2024–Present	IEEE Computer Society Member

REFERENCES

Dr. Raihan UI Islam Associate Professor Department of Computer Science & Engineering East West University, Bangladesh Tel. +880 1927 673124 raihan.islam@ewubd.edu	Nakib Aman Assistant Professor Department of Computer Science & Engineering Pabna University of Science and Technology Tel. +8801762910933 nakib.cse@pust.ac.bd
--	---